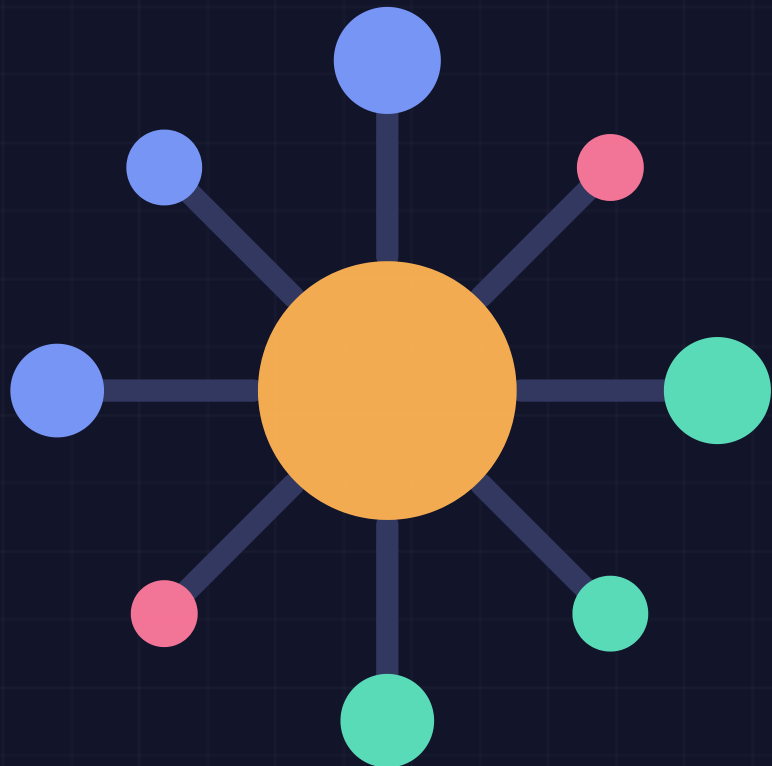


Neuron.

A practical AI course for curious kids

Twenty short lessons that teach a 10–15-year-old how to actually use AI: ask it well, know when to trust it, build things with it, and stay in charge of it. Works offline, keeps no account, and rewards every lesson with a piece of a hidden picture.



Neuron · repository [miekuskamil/ai-academy-game](#)

Build reviewed: branch [review-fixes](#), commit [474e92a](#)

26 September 2026 · document 1 of 3 (Overview · Architecture · Logic)

What Neuron is

Neuron is a web app that teaches practical AI skills to one child at a time, on a phone, tablet or laptop. It opens with the most useful skill first — writing a clear request — and only then explains what is happening inside the machine.

6 worlds

from “Talking to AI” to “AI that does things”

20 lessons

10–15 minutes each, in a fixed order

7 activities

hands-on puzzles inside 10 of the lessons

Who it is for

Children aged roughly 10 to 15. Two reading paces cover that range: **Explorer** (10–12) keeps more of the story and takes gentler steps; **Builder** (13–15) covers the same ideas with less hand-holding. A grown-up can switch pace at any time without losing progress.

What she can do at the end

- Write a request an AI can follow: who it is for, the job, the limits and the shape of the answer.
- Spot a confident but made-up answer, and decide how hard to check something before trusting it.
- Choose when AI is the right tool, when to do the job herself, and when to ask a grown-up.
- Describe an app to an AI, fix what comes back, and plan a real project such as a plant sensor.
- Give an AI helper a goal, watch its steps, and catch the wrong turn before it matters.

Design principles

OFFLINE FIRST

Works with no internet

The whole app is one self-contained file, or an installable web app that caches itself. Nothing is fetched while she learns.

PRIVATE BY DEFAULT

No account, nothing uploaded

Progress lives only on the device. A grown-up can save it to a file and load it on another device.

NEVER SHAMING

Mistakes are the lesson

Wrong answers get a kind nudge and an explanation. Retries never lower a score, and nothing is timed.

REWARDING

Every lesson earns something

Each finished lesson earns a jigsaw piece she places herself. Together they reveal what she was building all along.

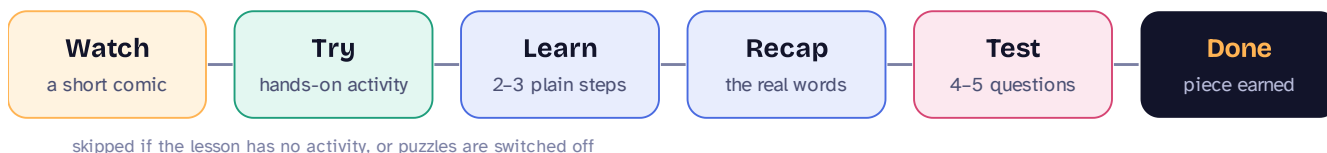
Six worlds, twenty lessons

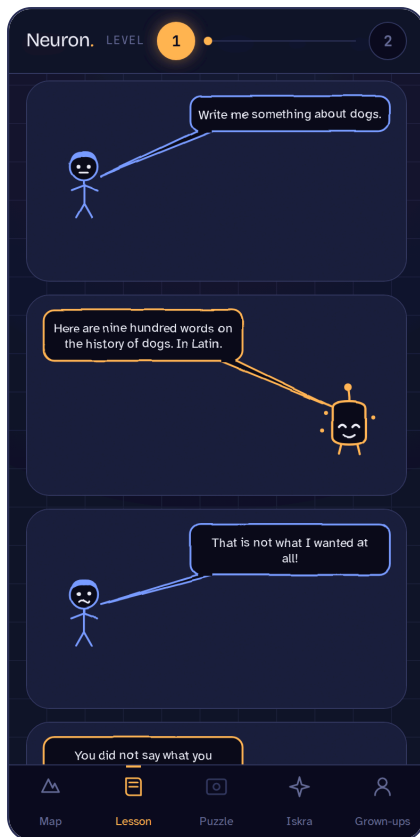
The order runs from doing to understanding: talk to AI first, then learn why it behaves as it does, then use it, build with it and put it to work.

WORLD	LESSONS	HANDS-ON ACTIVITY
1 • Talking to AI Ask well, and the answers get good.	Ask like you mean it · Show, don't just tell · One step at a time · Not quite — try again	Prompt lab (all four)
2 • Why it answers that way It is guessing the next word.	It guesses the next word · Why it makes things up · Same question, different answer	Spot the made-up answer · Temperature dial
3 • Using AI for real things When to trust it, check it, or do it yourself.	The right tool for the job · Trust, but check · A thinking partner · When not to use it	Task sorter · How hard to check
4 • What is going on inside It learned from a mountain of examples.	A mountain of examples · Patterns, not understanding · It picked up our habits	—
5 • Building with AI Describe it, and build it by talking.	Describe it and watch it build · Say what's wrong and fix it · AI in a real project · Make it yours	Pipeline builder
6 • AI that does things Give it a goal and tools, and stay the boss.	Give it a goal and tools · Watch it work, keep it honest	Keep the agent honest

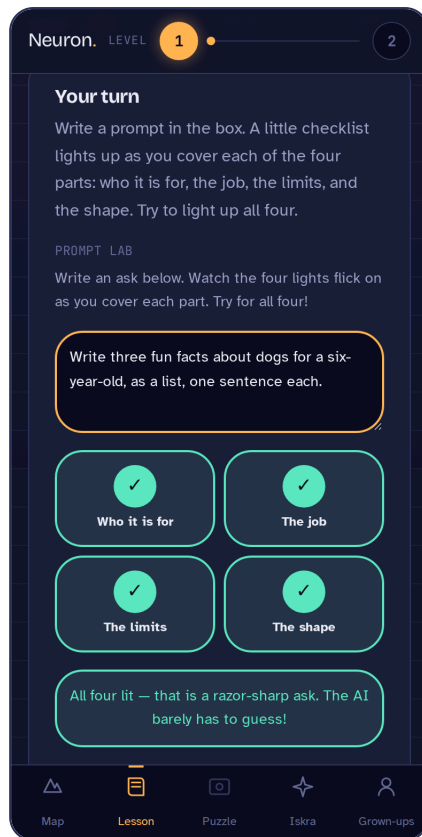
A lesson in six steps

Every lesson follows the same path, shown as a small progress line at the top. She can step back to any step she has already reached.

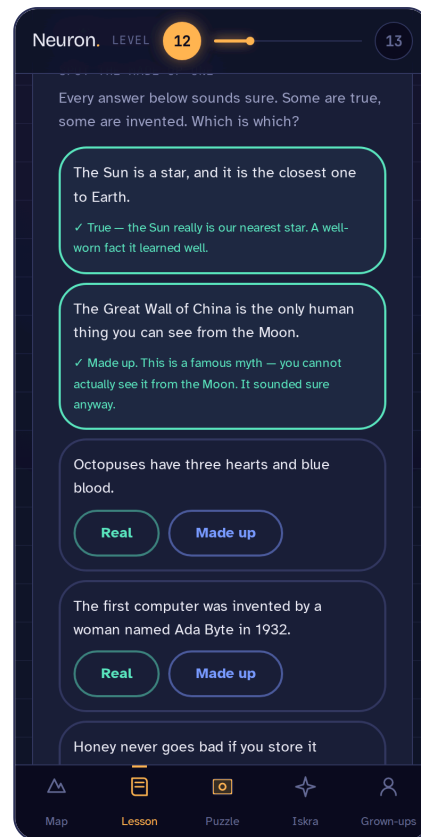




Watch — Iskra, the AI helper, makes a mistake the child can spot.



Try — the prompt lab lights up each part of a good request as she writes.



Try — judging which confident answers are true and which are made up.

03 • HANDS-ON

Seven activities

Each activity is played with taps, never drags, so it works reliably on a phone. Every one explains itself after each answer, and counts toward the lesson once she reaches its goal.

ACTIVITY	WHAT SHE DOES	GOAL TO PASS
Prompt lab World 1	Writes a request; four lights show whether it names who it is for, the job, the limits and the shape.	At least 3 of the 4 lights
Spot the made-up answer World 2	Reads eight confident answers and marks each real or made up, with the reason revealed each time.	At least 6 of 8 right
Temperature dial World 2	Moves a dial from steady to wild and asks again, watching the answers vary.	Try both ends of the dial
Task sorter World 3	Sorts eight real tasks into “ask AI”, “do it myself” or “check carefully”.	At least 6 of 8 in the best bin
How hard to check World 3	Decides whether each AI answer is fine to trust, needs checking, or needs a grown-up.	At least 4 of 6 matched

ACTIVITY	WHAT SHE DOES	GOAL TO PASS
Pipeline builder World 5	Puts the steps of an AI plant helper in order: sensor reads, AI decides, light blinks.	All steps in order
Keep the agent honest World 6	Reads three AI helper plans and taps the step that took a wrong turn.	At least 2 of 3 caught

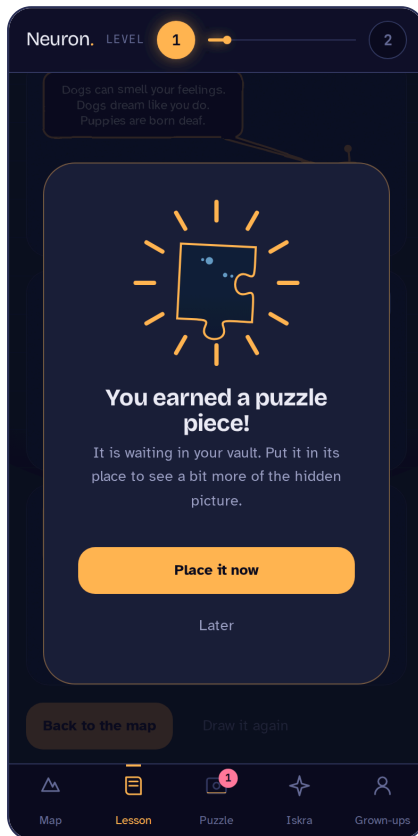
Questions in the Test step

Tests mix five question types: pick one or more answers, put items in order, match pairs, give a number, and write a request that is checked against a short list of must-haves. Questions shuffle on every attempt so nothing can be learned by position. A wrong answer shows why, and she can fix it straight away.

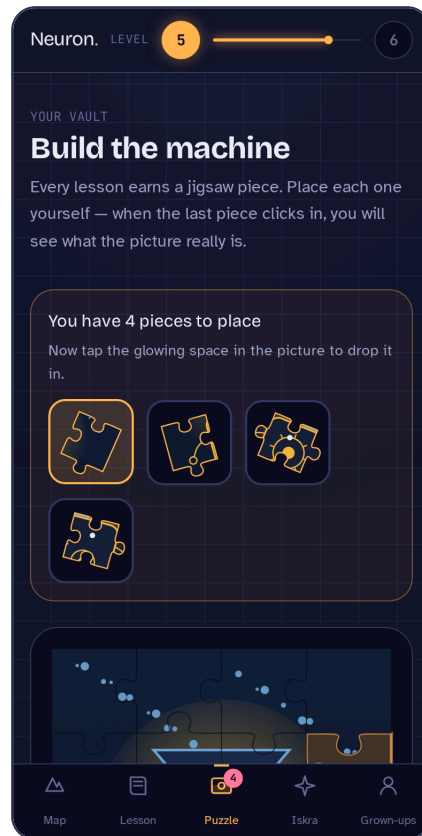
Passing a lesson. A lesson counts as finished at 60% of its points (or every question right). Finishing with 90% or more and no hints marks it **mastered**, shown as a star. She can come back any time to improve; a retry can only ever raise her best score.

The hidden picture

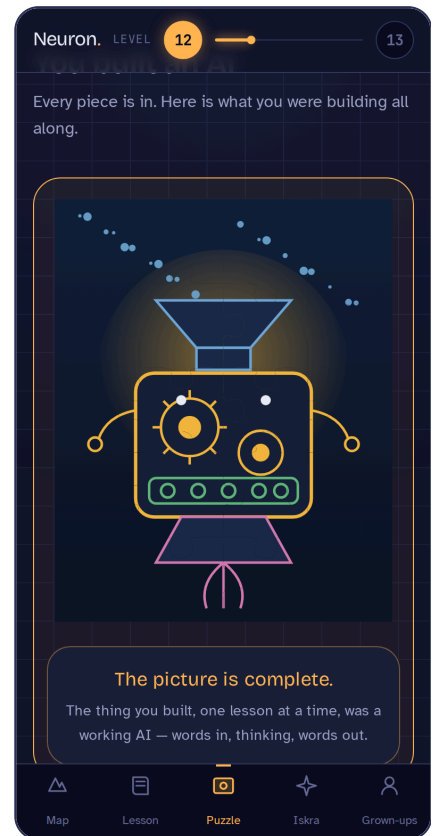
Every finished lesson earns one jigsaw piece. She places each piece herself, and the picture slowly reveals what she has been building: a friendly AI machine with words going in, thinking in the middle and words coming out.



The moment a lesson is finished: her own piece, and a button to go and place it.



Placing pieces: tap a piece, then the glowing space. Pieces are tilted so the picture stays a mystery.



All twenty placed: the picture is complete, and she can run the AI she built.

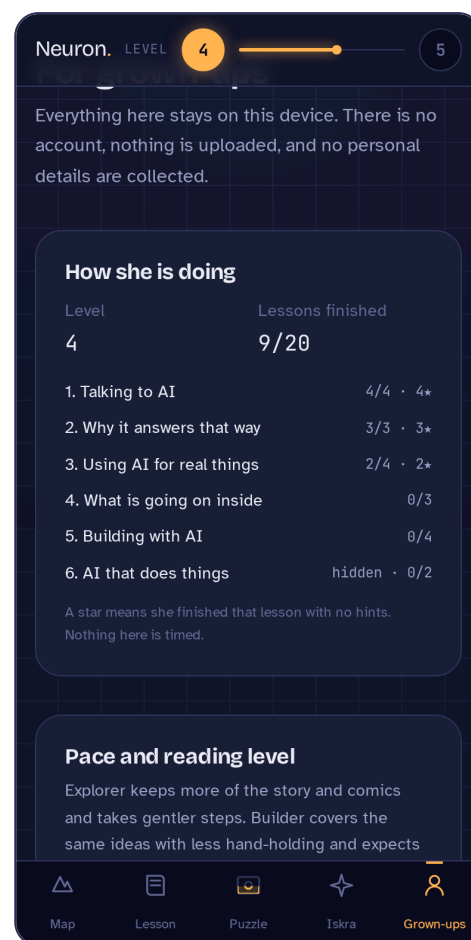
How it keeps her going

- **One reward, one place.** The jigsaw is the only reward during the course. The project she can run appears only once the picture is complete.
- **Never forgotten.** Pieces waiting to be placed show as a number on the Puzzle tab.
- **Hard to guess.** Empty spaces show only faint outlines, and waiting pieces sit tilted, so the picture only makes sense once several pieces join up.
- **Cannot be faked.** Pieces are worked out from finished lessons each time the page draws, so an edited save file cannot open the picture early.

Alongside the puzzle, a level counter at the top grows as she finishes lessons. Later lessons are worth more, so levels track how far she has come, not how many boxes she ticked.

Controls and privacy

CONTROL	WHAT IT DOES
How she is doing	Level, lessons finished, and progress per world with stars for no-hint finishes.
Pace	Explorer (10–12) or Builder (13–15). Switch any time.
Hands-on puzzles	Full; Gentle (any real try counts, so she is never stuck); or Skip (lesson scored on questions only).
Which worlds	Hide a world for later. Later worlds still open in order; progress is never deleted.
Save or move	Save progress to a file, or load it on another device.
Lock this page	Optional 4-digit code so only a grown-up can change these settings.
Movement	Follow the device, always animate, or no movement.
Start over	Erase all progress on this device, with a confirm step.



The grown-ups page, with World 6 hidden.

Privacy. There is no account, no sign-in and no analytics. Progress is stored only in the browser on the device, and the app makes no network requests while she learns. The code includes an adapter for a local AI model (Ollama) on a home network, but this build does not switch it on; every lesson works without one. The lessons teach her never to paste private details — addresses, passwords, faces, school records — into public AI tools.

Two ways to open it

SHARE A FILE

neuron.html

One self-contained file, about 750 KB. Open it in any modern browser, even in airplane mode. Best for a quick try; some messaging apps open files in a way that does not keep progress between visits.

INSTALL IT

neuron-pwa-site.zip

The installable web app. Drop it on Netlify to get a web address, open that address on the phone and choose “Add to Home screen”, or feed the address to PWABuilder to make an Android app. Progress is kept reliably.

Updating the repository

The review fixes come as one patch. From the project folder on the Mac:

```
git am ~/Downloads/neuron-fixes.patch && git push
```

Then run `npm install && npm run build` to rebuild. The offline cache is versioned (`neuron-v2`), so phones that already installed the app pick up the new version on their next visit.

Current state

CHECK	RESULT
App tests (Vitest)	222 passing
Content pipeline tests (pytest)	64 passing
Type check, lint and layer rules	clean
Every style class produces CSS	checked on each build
Script size (gzip)	113.9 KB of 200 KB budget
Hardest lesson reading level	grade 5.2
Browser walk at phone and desktop size	no errors, no sideways scroll

For how the parts fit together, see **Architecture**. For the exact rules behind scoring, unlocking and rewards, see **Logic**.